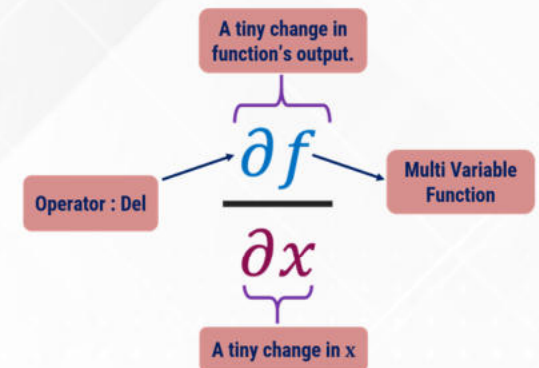




Unit-4

Partial Derivatives



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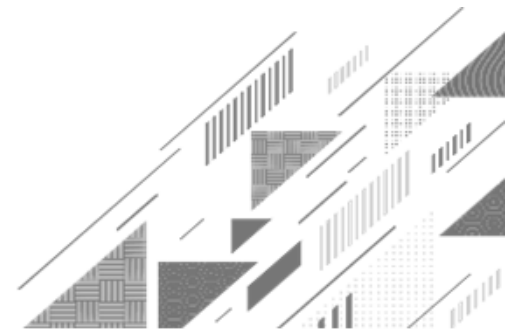
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OUTLINE

- Method:1 ➡ Limit of Function of Two Variables
- Method:2 ➡ Continuity of Function of Two Variables
- Method:3 ➡ Partial Derivatives
- Method:4 ➡ Chain Rule
- Method:5 ➡ Implicit Function
- Method:6 ➡ Tangent Plane and Normal Line



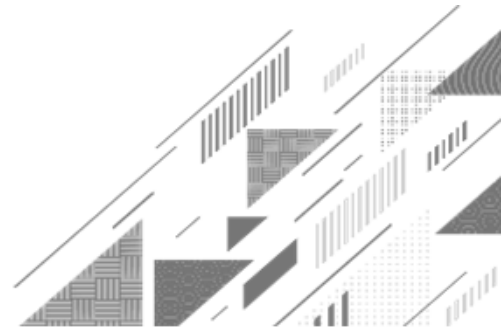


OUTLINE

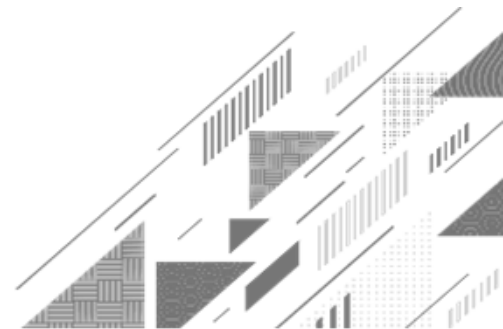


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- Method:7 ➡ Local Extreme Values
- Method:8 ➡ Lagrange's Multipliers
- Method:9 ➡ Gradient
- Method:10 ➡ Directional Derivative



Method:1 ➡ Limit of Function of Two Variables

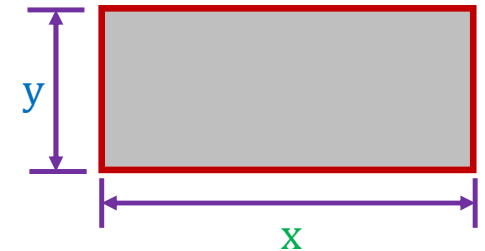


Function of Several Variables

► A function of **two** or **more than two** **variables** is said to be function of several variables.

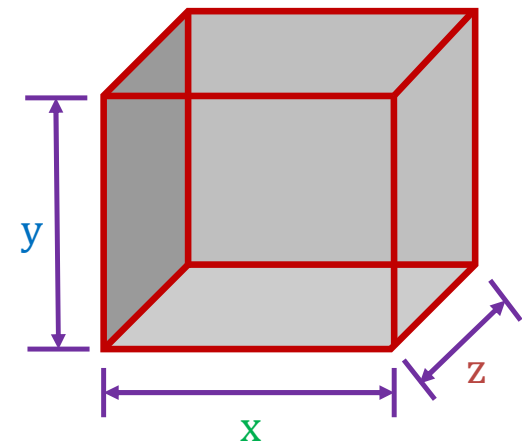
→ The **area** of **rectangle** is a function of two variables.

$$A = f(x, y) = x \cdot y$$



→ The **volume** of a **rectangular box** is a function of three variables.

$$V = f(x, y, z) = x \cdot y \cdot z$$



Example of Limit of Function of Two Variables $f(x, y)$

Example 1: Evaluate $\lim_{(x, y) \rightarrow (5, -2)} x^4 + 4x^3y - 5xy^2$.

Solution:

→ $\lim_{(x, y) \rightarrow (5, -2)} x^4 + 4x^3y - 5xy^2$ Substitute $x = 5$ & $y = -2$.

$$= (5)^4 + 4(5)^3(-2) - 5(5)(-2)^2$$

$$= 625 - 1000 - 100$$

$$= -475$$

Limit of a function $f(x, y)$ at point $(0, 0)$

► Procedure to find $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$.

(1). Find $L_1 = \lim_{y \rightarrow 0} \left\{ \lim_{x \rightarrow 0} f(x, y) \right\}$

(2). Find $L_2 = \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow 0} f(x, y) \right\}$

(3). Find $L_3 = \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx} f(x, y) \right\}$

(4). Find $L_4 = \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx^2} f(x, y) \right\}$

➡ If $L_1 = L_2 = L_3 = L_4 = L$ then limit is exist and $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = L$.

➡ If one of them is **not equal** to others then limit does **not exist**.

Example of Limit of Function of Two Variables at (0, 0)

Example 2: Find $\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2}$ if it exist.

$$f(x, y) = \frac{3x^2y}{x^2 + y^2}$$

Solution:

$$\begin{aligned}(1). L_1 &= \lim_{y \rightarrow 0} \left\{ \lim_{x \rightarrow 0} f(x, y) \right\} \\&= \lim_{y \rightarrow 0} \left\{ \lim_{x \rightarrow 0} \frac{3x^2y}{x^2 + y^2} \right\} \\&= \lim_{y \rightarrow 0} \left\{ \frac{3(0)^2y}{0^2 + y^2} \right\} \\&= \lim_{y \rightarrow 0} \left\{ \frac{0}{y^2} \right\} = \lim_{y \rightarrow 0} \{0\}\end{aligned}$$

$$\Rightarrow \boxed{L_1 = 0}$$

$$\begin{aligned}(2). L_2 &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow 0} f(x, y) \right\} \\&= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow 0} \frac{3x^2y}{x^2 + y^2} \right\} \\&= \lim_{x \rightarrow 0} \left\{ \frac{3x^2(0)}{x^2 + 0^2} \right\} \\&= \lim_{x \rightarrow 0} \left\{ \frac{0}{x^2} \right\} = \lim_{x \rightarrow 0} \{0\}\end{aligned}$$

$$\Rightarrow \boxed{L_2 = 0}$$

Example of Limit of Function of Two Variables at (0, 0)

Example 2: Find $\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2}$ if it exist.

$$f(x, y) = \frac{3x^2y}{x^2 + y^2}$$

Solution Continue:

$$\begin{aligned}(3). L_3 &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx} f(x, y) \right\} \\&= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx} \frac{3x^2y}{x^2 + y^2} \right\} \\&= \lim_{x \rightarrow 0} \left\{ \frac{3x^2 \cdot mx}{x^2 + (mx)^2} \right\} \\&= \lim_{x \rightarrow 0} \left\{ \frac{3mx^{\cancel{2}}}{\cancel{x^2}(1 + m^2)} \right\}\end{aligned}$$

$$\begin{aligned}&= \lim_{x \rightarrow 0} \left\{ \frac{3mx}{1 + m^2} \right\} \\&= \frac{3m(0)}{1 + m^2} \\&= \frac{0}{1 + m^2} = 0 \\&\Rightarrow \boxed{L_3 = 0}\end{aligned}$$

Example of Limit of Function of Two Variables at (0, 0)

Example 2: Find $\lim_{(x,y) \rightarrow (0,0)} \frac{3x^2y}{x^2 + y^2}$ if it exist.

$$f(x, y) = \frac{3x^2y}{x^2 + y^2}$$

Solution Continue:

$$\begin{aligned}(4). L_4 &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx^2} f(x, y) \right\} \\&= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx^2} \frac{3x^2y}{x^2 + y^2} \right\} \\&= \lim_{x \rightarrow 0} \left\{ \frac{3x^2 \cdot mx^2}{x^2 + (mx^2)^2} \right\} \\&= \lim_{x \rightarrow 0} \left\{ \frac{3mx^4}{\cancel{x^2}(1 + m^2x^2)} \right\}\end{aligned}$$

$$\begin{aligned}&= \lim_{x \rightarrow 0} \left\{ \frac{3mx^2}{1 + m^2x^2} \right\} \\&= \frac{3m(0)^2}{1 + m^2(0)^2} \\&= \frac{0}{1} = 0 \\&\Rightarrow \boxed{L_4 = 0}\end{aligned}$$

Example of Limit of Function of Two Variables at (0, 0)

Example 2: Find $\lim_{(x, y) \rightarrow (0, 0)} \frac{3x^2y}{x^2 + y^2}$ if it exist.

Solution Continue:

Here, $L_1 = L_2 = L_3 = L_4 = L = 0$.

$$\Rightarrow \lim_{(x, y) \rightarrow (0, 0)} \frac{3x^2y}{x^2 + y^2} = 0$$

Example of Limit of Function of Two Variables at (0, 0)

Example 5: Show that $f(x, y) = \frac{2x^2y}{x^4 + y^2}$ has no limit as (x, y) approaches to $(0, 0)$.

Solution:

$$\begin{aligned}(1). L_1 &= \lim_{y \rightarrow 0} \left\{ \lim_{x \rightarrow 0} f(x, y) \right\} \\&= \lim_{y \rightarrow 0} \left\{ \lim_{x \rightarrow 0} \frac{2x^2y}{x^4 + y^2} \right\} \\&= \lim_{y \rightarrow 0} \left\{ \frac{0}{y^2} \right\} \\&= \lim_{y \rightarrow 0} \{0\} \\&= 0 \\&\Rightarrow \boxed{L_1 = 0}\end{aligned}$$

$$\begin{aligned}(2). L_2 &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow 0} f(x, y) \right\} \\&= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow 0} \frac{2x^2y}{x^4 + y^2} \right\} \\&= \lim_{x \rightarrow 0} \left\{ \frac{0}{x^4} \right\} \\&= \lim_{x \rightarrow 0} \{0\} \\&= 0 \\&\Rightarrow \boxed{L_2 = 0}\end{aligned}$$

Example of Limit of Function of Two Variables at (0, 0)

Example 5: Show that $f(x, y) = \frac{2x^2y}{x^4 + y^2}$ has no limit as (x, y) approaches to $(0, 0)$.

Solution Continue:

$$\begin{aligned}(3). L_3 &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx} f(x, y) \right\} \\&= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx} \frac{2x^2y}{x^4 + y^2} \right\} \\&= \lim_{x \rightarrow 0} \left\{ \frac{2x^2 \cdot mx}{x^4 + (mx)^2} \right\} \\&= \lim_{x \rightarrow 0} \left\{ \frac{2mx^3}{\cancel{x^2}(x^2 + m^2)} \right\}\end{aligned}$$

$$\begin{aligned}&= \lim_{x \rightarrow 0} \left\{ \frac{2mx}{x^2 + m^2} \right\} \\&= \frac{0}{m^2} \\&= 0 \\&\Rightarrow \boxed{L_3 = 0}\end{aligned}$$

Example of Limit of Function of Two Variables at (0, 0)

Example 5: Show that $f(x, y) = \frac{2x^2y}{x^4 + y^2}$ has no limit as (x, y) approaches to $(0, 0)$.

Solution Continue:

$$\begin{aligned}(4). L_4 &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx^2} f(x, y) \right\} \\&= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx^2} \frac{2x^2y}{x^4 + y^2} \right\} \\&= \lim_{x \rightarrow 0} \left\{ \frac{2x^2 \cdot mx^2}{x^4 + (mx^2)^2} \right\} \\&= \lim_{x \rightarrow 0} \left\{ \frac{2mx^{\cancel{4}}}{\cancel{x^4}(1 + m^2)} \right\}\end{aligned}$$

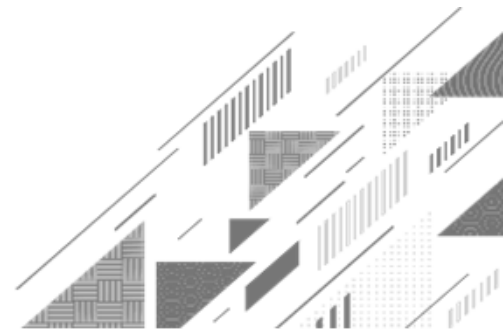
$$= \lim_{x \rightarrow 0} \left\{ \frac{2m}{1 + m^2} \right\}$$

$$= \frac{2m}{1 + m^2} \Rightarrow L_4 \text{ depends on } m.$$

Hence, $L_1 = L_2 = L_3 \neq L_4$.

$$\Rightarrow \lim_{(x, y) \rightarrow (0, 0)} \frac{2x^2y}{x^4 + y^2} \text{ does not exist.}$$

Method:2 ➡ Continuity of Function of Two Variables



Continuity of Function of Two Variables $f(x, y)$ at $(0, 0)$

► Function $f(x, y)$ is said to be **continuous** at point $(0, 0)$ if the following conditions are satisfied.

(1). $f(0, 0)$ must be **well define**.

(2). $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ must be **exist**.

(3). $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = f(0, 0)$.

(1). Find $L_1 = \lim_{y \rightarrow 0} \left\{ \lim_{x \rightarrow 0} f(x, y) \right\}$

(2). Find $L_2 = \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow 0} f(x, y) \right\}$

(3). Find $L_3 = \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx} f(x, y) \right\}$

(4). Find $L_4 = \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx^2} f(x, y) \right\}$

→ $L_1 = L_2 = L_3 = L_4 = L$

Note: If **one** of the above three condition is **not satisfied** then $f(x, y)$ is **not continuous** at point $(0, 0)$.

Example of Continuity of Function of Two Variables at (0, 0)

Example 1: Discuss the continuity of the function f defined as

$$f(x, y) = \begin{cases} \frac{x^3 - y^3}{x^2 + y^2} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}.$$

Solution:

Here, $f(0, 0) = 0$.

Condition 1: $f(0, 0)$ must be well define.

$\Rightarrow$ f is well defined at $(0, 0)$

Example of Continuity of Function of Two Variables at (0, 0)

Example 1: Discuss the continuity of the function f defined as

$$f(x, y) = \begin{cases} \frac{x^3 - y^3}{x^2 + y^2} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}.$$

Solution Continue:

Condition 2: $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ must be exist.

$$= \lim_{y \rightarrow 0} \left\{ \lim_{x \rightarrow 0} \frac{x^3 - y^3}{x^2 + y^2} \right\}$$

$$= \lim_{y \rightarrow 0} \left\{ \frac{-y^3}{y^2} \right\}$$

$$= \lim_{y \rightarrow 0} \{-y\}$$

$$= 0$$

$$\Rightarrow \boxed{L_1 = 0}$$

Example of Continuity of Function of Two Variables at (0, 0)

Example 1: Discuss the continuity of the function f defined as

$$f(x, y) = \begin{cases} \frac{x^3 - y^3}{x^2 + y^2} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}.$$

Solution Continue:

$$\begin{aligned} (2). L_2 &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow 0} f(x, y) \right\} \\ &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow 0} \frac{x^3 - y^3}{x^2 + y^2} \right\} \\ &= \lim_{x \rightarrow 0} \left\{ \frac{x^{\cancel{3}}}{x^{\cancel{2}}} \right\} \end{aligned}$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \{x\} \\ &= 0 \\ \Rightarrow L_2 &= 0 \end{aligned}$$

Example of Continuity of Function of Two Variables at (0, 0)

Example 1: Discuss the continuity of the function f defined as

$$f(x, y) = \begin{cases} \frac{x^3 - y^3}{x^2 + y^2} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}.$$

Solution Continue:

$$\begin{aligned} (3). L_3 &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx} f(x, y) \right\} \\ &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx} \frac{x^3 - y^3}{x^2 + y^2} \right\} \\ &= \lim_{x \rightarrow 0} \left\{ \frac{x^3 - (mx)^3}{x^2 + (mx)^2} \right\} \end{aligned}$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{\cancel{x^3}(1 - m^3)}{\cancel{x^2}(1 + m^2)} \\ &= \lim_{x \rightarrow 0} \frac{x(1 - m^3)}{(1 + m^2)} \\ &= 0 \\ &\Rightarrow \boxed{L_3 = 0} \end{aligned}$$

Example of Continuity of Function of Two Variables at (0, 0)

Example 1: Discuss the continuity of the function f defined as

$$f(x, y) = \begin{cases} \frac{x^3 - y^3}{x^2 + y^2} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}.$$

Solution Continue:

$$\begin{aligned} (4). L_4 &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx^2} f(x, y) \right\} \\ &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx^2} \frac{x^3 - y^3}{x^2 + y^2} \right\} \\ &= \lim_{x \rightarrow 0} \left\{ \frac{x^3 - (mx^2)^3}{x^2 + (mx^2)^2} \right\} \end{aligned}$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{x^{\cancel{3}}(1 - m^3x^3)}{\cancel{x^2}(1 + m^2x^2)} \\ &= \lim_{x \rightarrow 0} \frac{x(1 - m^3x^3)}{(1 + m^2x^2)} \\ &= 0 \\ &\Rightarrow \boxed{L_4 = 0} \end{aligned}$$

Example of Continuity of Function of Two Variables at (0, 0)

Example 1: Discuss the continuity of the function f defined as

$$f(x, y) = \begin{cases} \frac{x^3 - y^3}{x^2 + y^2} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}.$$

Solution Continue:

Here, $L_1 = L_2 = L_3 = L_4 = L = 0$.

$$\Rightarrow \lim_{(x, y) \rightarrow (0, 0)} \frac{x^3 - y^3}{x^2 + y^2} = 0$$

Condition 3: $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = f(0, 0)$.

Hence, all three conditions are satisfied.

$\Rightarrow f(x, y)$ is continuous at $(0, 0)$

Example of Continuity of Function of Two Variables at (0, 0)

Example 1*: Discuss the continuity of the function f defined as

$$f(x, y) = \begin{cases} \frac{x^3 - y^3}{x^2 + y^2} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}$$

Solution: 1: $f(0, 0)$ must be well define. ✓ $\because f(0, 0) = 2$

2: $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ must be exist. ✓ $\because \lim_{(x, y) \rightarrow (0, 0)} f(x, y) = 0$

3: $\lim_{(x, y) \rightarrow (0, 0)} f(x, y) = f(0, 0)$. ✗ $\because 0 \neq 2$

Hence, third condition is not satisfied.

$\Rightarrow$ **$f(x, y)$ is not continuous at $(0, 0)$.**

Example of Continuity of Function of Two Variables at (0, 0)

Example 5: Show that the following function is not continuous at origin.

$$f(x, y) = \begin{cases} \frac{2x^2y}{x^3 + y^3} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}.$$

Solution:

Here, $f(0, 0) = 0$.

Condition 1: $f(0, 0)$ must be well define.

$\therefore$ f is well defined at $(0, 0)$.

Example of Continuity of Function of Two Variables at (0, 0)

Example 5: Show that the following function is not continuous at origin.

$$f(x, y) = \begin{cases} \frac{2x^2y}{x^3 + y^3} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}$$

Solution Continue:

Condition 2: $\lim_{(x,y) \rightarrow (0,0)} \lim_{y \rightarrow 0} f(x, y)$ must be exist.

$$\begin{aligned} (1). L_1 &= \lim_{y \rightarrow 0} \left\{ \lim_{x \rightarrow 0} \frac{2x^2y}{x^3 + y^3} \right\} \\ &= \lim_{y \rightarrow 0} \left\{ \frac{0}{y^3} \right\} \\ &= 0 \\ \Rightarrow L_1 &= 0 \end{aligned}$$

Example of Continuity of Function of Two Variables at (0, 0)

Example 5: Show that the following function is not continuous at origin.

$$f(x, y) = \begin{cases} \frac{2x^2y}{x^3 + y^3} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}$$

Solution Continue:

$$\begin{aligned} (2). L_2 &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow 0} f(x, y) \right\} \\ &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow 0} \frac{2x^2y}{x^3 + y^3} \right\} \\ &= \lim_{x \rightarrow 0} \left\{ \frac{0}{x^3} \right\} \end{aligned}$$

$$\begin{aligned} &= 0 \\ \Rightarrow L_2 &= 0 \end{aligned}$$

Example of Continuity of Function of Two Variables at (0, 0)

Example 5: Show that the following function is not continuous at origin.

$$f(x, y) = \begin{cases} \frac{2x^2y}{x^3 + y^3} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}$$

Solution Continue:

$$\begin{aligned} (3). L_3 &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx} f(x, y) \right\} \\ &= \lim_{x \rightarrow 0} \left\{ \lim_{y \rightarrow mx} \frac{2x^2y}{x^3 + y^3} \right\} \\ &= \lim_{x \rightarrow 0} \left\{ \frac{2x^2 \cdot mx}{x^3 + m^3x^3} \right\} \end{aligned}$$

$$\begin{aligned} &= \lim_{x \rightarrow 0} \frac{2\cancel{m}x^{\cancel{3}}}{\cancel{x^3}(1 + m^3)} \\ &= \frac{2m}{1 + m^3} \quad (\text{Which depends on } m) \\ &\text{Hence, } L_1 = L_2 \neq L_3. \end{aligned}$$

Example of Continuity of Function of Two Variables at (0, 0)

Example 5: Show that the following function is not continuous at origin.

$$f(x, y) = \begin{cases} \frac{2x^2y}{x^3 + y^3} & ; (x, y) \neq (0, 0) \\ 0 & ; (x, y) = (0, 0) \end{cases}$$

Solution Continue:

$$\Rightarrow \lim_{(x, y) \rightarrow (0, 0)} \frac{2x^2y}{x^3 + y^3} \text{ does not exist.}$$

Here, second condition is not satisfied.

$$\Rightarrow f(x, y) \text{ is not continuous at } (0, 0).$$

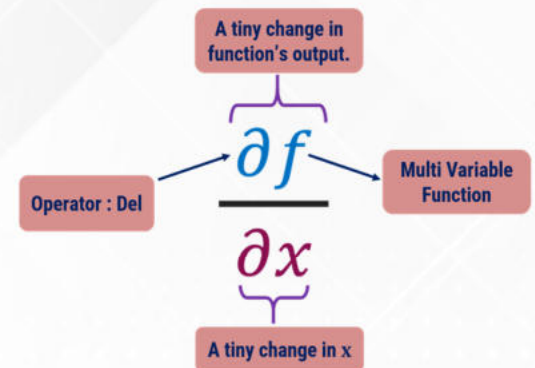
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